just like an energetic point particle. But when its energy is increased beyond the value required for probing structures on the scale of the Planck length, the additional energy does not sharpen the string probe. Rather, the energy causes the string to grow in size, thereby diminishing its short-distance sensitivity. In fact, although the size of a typical string is the Planck length, if we pumped enough energy into a string—an amount of energy beyond our wildest imaginings but one that would likely have been attained by the big bang—we could cause it to grow to a macroscopic size, a clumsy probe of the microcosmos indeed! It's as if a string, unlike a point particle, has two sources of smearing: quantum jitters, as for a point particle, and also its own inherent spatial extent. Increasing a string's energy decreases the smearing from the first source but ultimately increases the smearing from the second. The upshot is that no matter how hard you try, the extended nature of a string prevents you from using it to probe phenomena on sub-Planck-length distances.

But the whole conflict between general relativity and quantum mechanics arises from the sub-Planck-length properties of the spatial fabric. If the elementary constituent of the universe cannot probe sub-Planck-scale distances, then neither it nor anything made from it can be affected by the supposedly disastrous short-distance quantum undulations. This is similar to what happens as we draw our hand across a highly polished granite surface. Although at a microscopic level the granite is discrete, grainy, and bumpy, our fingers are unable to detect these short-scale variations and the surface feels perfectly smooth. Our stumpy, extended fingers "smear" out the microscopic discreteness. Similarly, since the string has spatial extent, it also has limits on its short-distance sensitivity. It cannot detect variations on sub-Planck-distance scales. Like our fingers on granite, the string smears out the jittery ultramicroscopic fluctuations of the gravitational field. Although the resulting fluctuations are still substantial, this smearing smooths them out just enough to cure the incompatibility between general relativity and quantum mechanics. And, in particular, the pernicious infinities (discussed in the preceding chapter) that arise in the point-particle approach to forming a quantum theory of gravity are done away with by string theory.